

Physical Chemistry (I)

Lecture 15

Simple Mixtures



In the Last Class

- van der Waals Fluid
 - Liquid-gas transition
 - Supercritical fluid
 - Critical point
- Thermodynamics of phase coexistence
 - T - and p -dependence of chemical potential
 - Shapes of phase boundaries

Last class

Good Old Friend

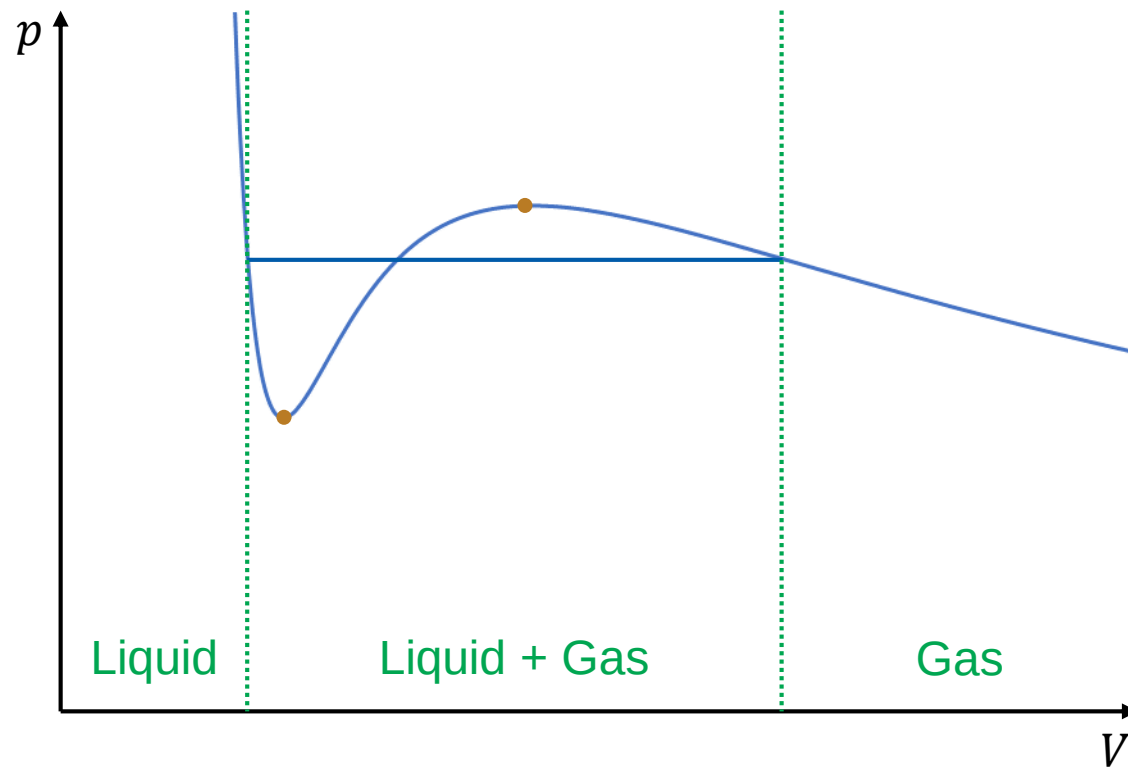
“Can we describe the liquid-gas transition using a model system?”

The van der Waals fluid!

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

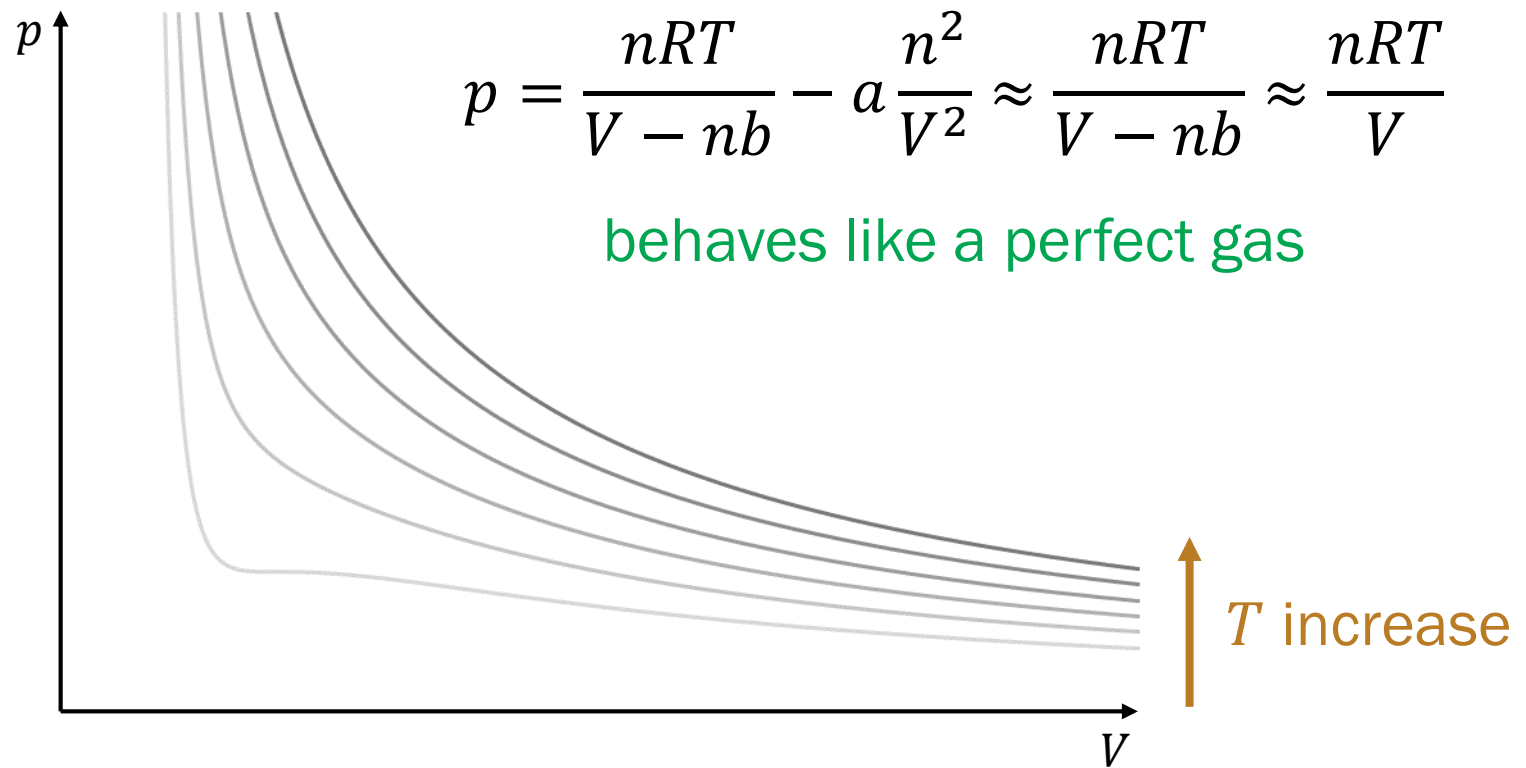
Last class

Low T : Liquid-Gas Transition

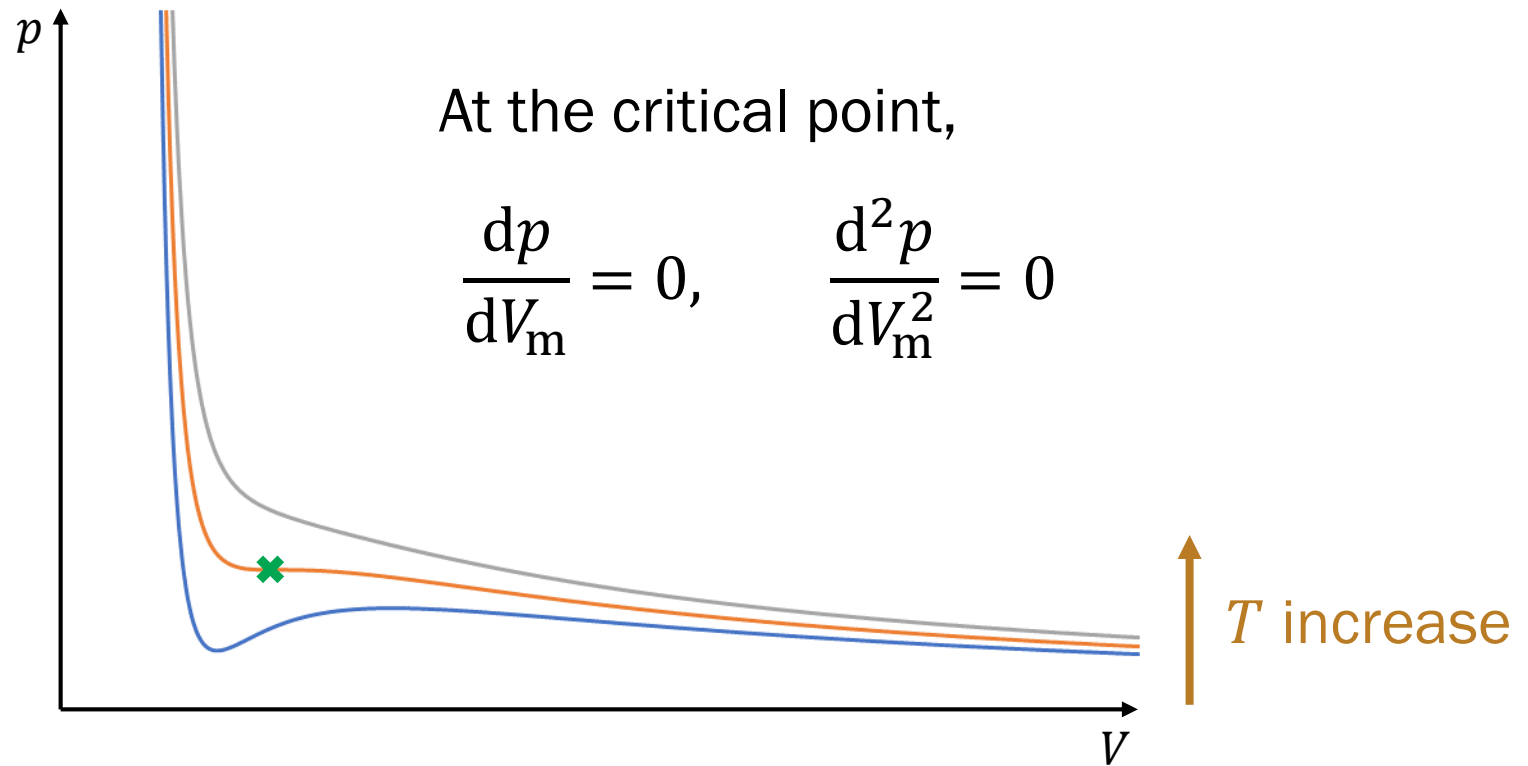


Last class

High T : Supercritical Fluid



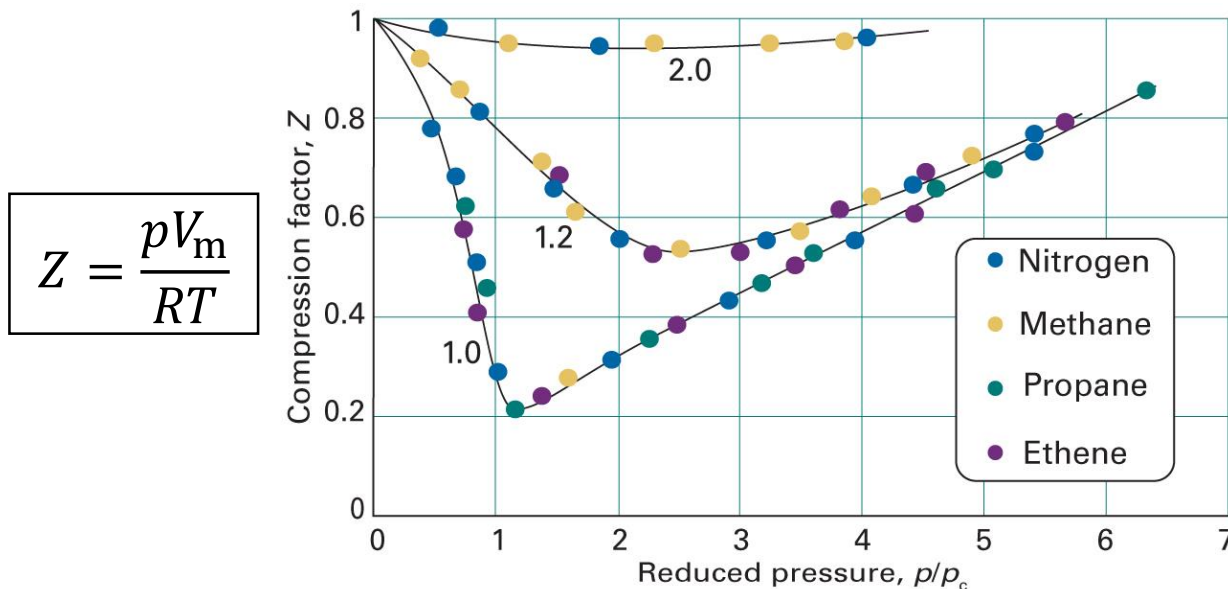
Critical Point



Last class

Principle of Corresponding States

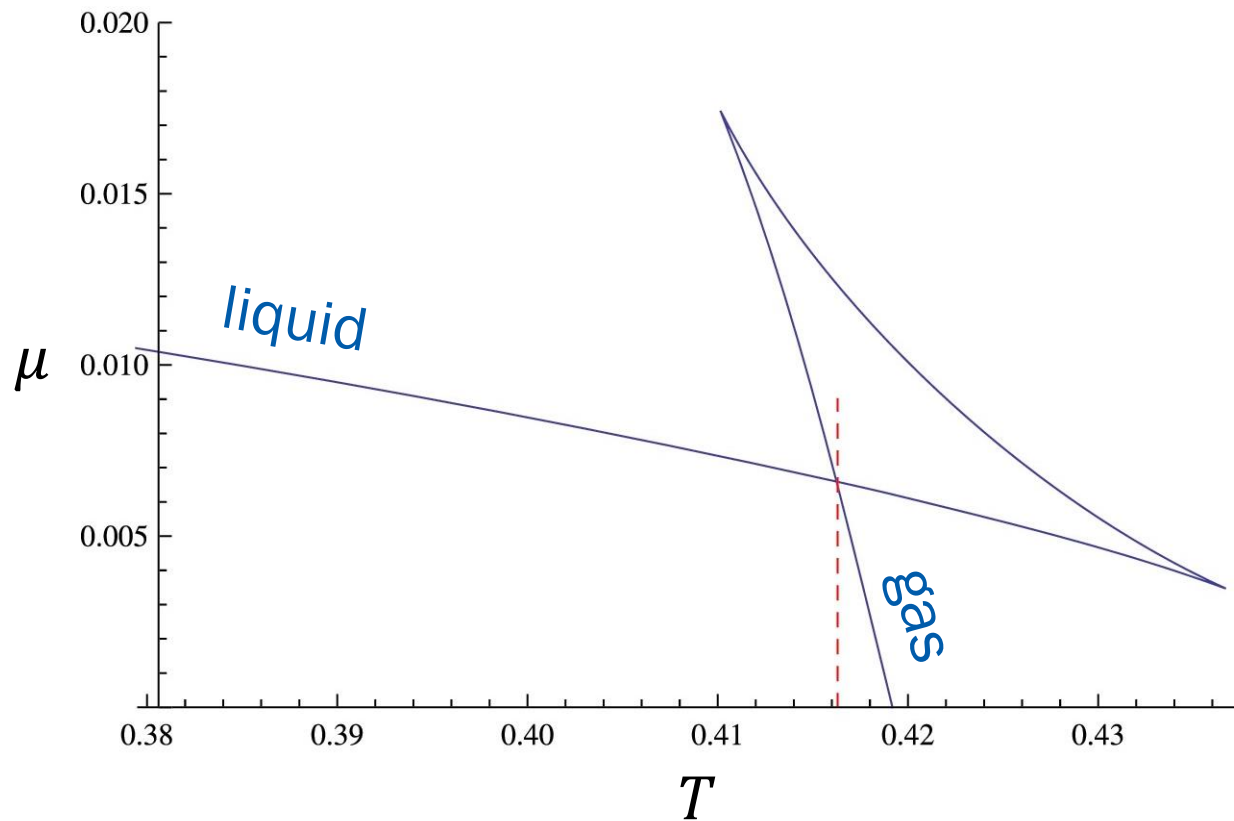
Gases at the same V_r and T_r exert the same p_r



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figures 10.9)

Last class

$\mu(T)$ of vdW Fluid



(Image: He et al., *Nuclear Physics B* 915: 243-261 (2017), Figure 3)

Last class

Chemical Potential

Recall

$$d\mu = -S_m dT + V_m dp$$

which gives

$$\left(\frac{\partial \mu}{\partial T}\right)_{p,n} = -S_m, \quad \left(\frac{\partial \mu}{\partial p}\right)_{T,n} = V_m$$

Clapeyron Equation

When two phases α and β are at equilibrium,

$$\mu(\alpha; p, T) = \mu(\beta; p, T)$$

Using $d\mu = V_m dp - S_m dT$ (Gibbs-Duhem equation), one can show

$$\frac{dp}{dT} = \frac{\Delta_{\text{trs}} S}{\Delta_{\text{trs}} V}, \quad \frac{dT}{dp} = \frac{\Delta_{\text{trs}} V}{\Delta_{\text{trs}} S}$$

Solid-Liquid Boundary

$$\frac{dp}{dT} = \frac{\Delta_{\text{fus}}H}{T\Delta_{\text{fus}}V}$$

Assuming $\Delta_{\text{fus}}H$ and $\Delta_{\text{fus}}V$ are constant, one can show

$$p_2 - p_1 = \frac{\Delta_{\text{fus}}H}{T_1\Delta_{\text{fus}}V} (T_2 - T_1)$$

Liquid-Gas Boundary

$$\frac{dp}{dT} = \frac{\Delta_{\text{vap}}H}{T\Delta_{\text{vap}}V}$$

Since $V_{\text{m}}(g) \gg V_{\text{m}}(l)$,

$$\Delta_{\text{vap}}V = V_{\text{m}}(g) - V_{\text{m}}(l) \approx V_{\text{m}}(g) \approx RT/p$$

Thus,

$$\frac{1}{p} \frac{dp}{dT} = \frac{\Delta_{\text{vap}}H}{RT^2}$$

(Clausius-Clapeyron equation)

Solid-Gas vs. Liquid-Gas

$$\text{Solid-gas: } \ln\left(\frac{p_2}{p_1}\right) = \frac{\Delta_{\text{sub}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\text{Liquid-gas: } \ln\left(\frac{p_2}{p_1}\right) = \frac{\Delta_{\text{vap}}H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

Since $\Delta_{\text{sub}}H = \Delta_{\text{fus}}H + \Delta_{\text{vap}}H > \Delta_{\text{vap}}H$, the sublimation curve has a steeper slope than the vaporization curve

8. Simple Mixtures

Chapter Overview

- Multi-component systems
- Mixing of gases
 - Mixing of perfect gases
 - Mixing of real gases
- Mixing of liquids and solids
 - Ideal solutions
 - Real solutions
 - Liquid-solid solutions: colligative properties



Multi-Component System

- State: $(p, T, n) \rightarrow (p, T, n_A, n_B, \dots)$
- Thermodynamic potentials
 - $dU = TdS - pdV + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dH = TdS + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dA = -SdT - pdV + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dG = -SdT + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$

Multi-Component System

- Possible phases
 - Each component: gas, liquid, solid
 - Mixed/demixed
- e.g. two-component system (binary system)
 - A: liquid, B: liquid, A-B: mixed
 - A: liquid, B: liquid, A-B: demixed
 - A: gas, B: gas, A-B: mixed
 - ...

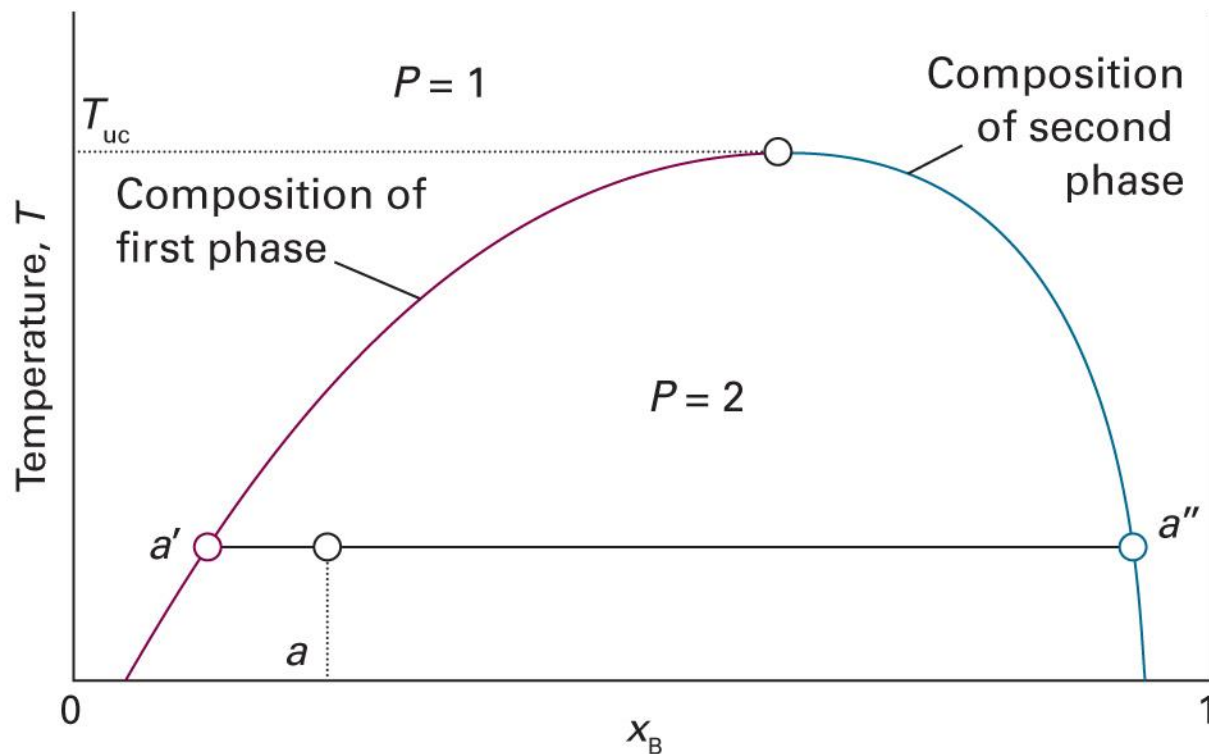
Phase Diagram

- Mole fraction of species J

$$x_J = \frac{n_J}{\sum_J n_J}$$

- At least one axis (usually the x-axis) is for x_J
- The other is for $T, p, \dots$

Example: Binary System



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5C.14)

Phase Rule

- Phase rule: a general relation between
 - Number of degrees of freedom F
 - Number of components C
 - Number of phases at equilibrium P
 - $F = (\text{\# of independent variables}) - (\text{\# of restrictions})$
- Criteria for phase stability (same μ)
 - For each component J , μ_J must be identical
 - $\mu_J(\text{phase 1}; p, T) = \mu_J(\text{phase 2}; p, T) = \dots$

Generalized Phase Rule

- The total number of independent intensive variables
 - T and p : 2
 - $(C - 1)$ mole fractions (-1 due to normalization) for each phase: $P(C - 1)$
 - Total number = $P(C - 1) + 2$
- The total number of equations (restrictions)
 - $\mu_J(\text{phase 1}; p, T) = \mu_J(\text{phase 2}; p, T) = \dots$ for component J
 - Total number = $C(P - 1)$
- Therefore, $F = P(C - 1) + 2 - C(P - 1) = C - P + 2$



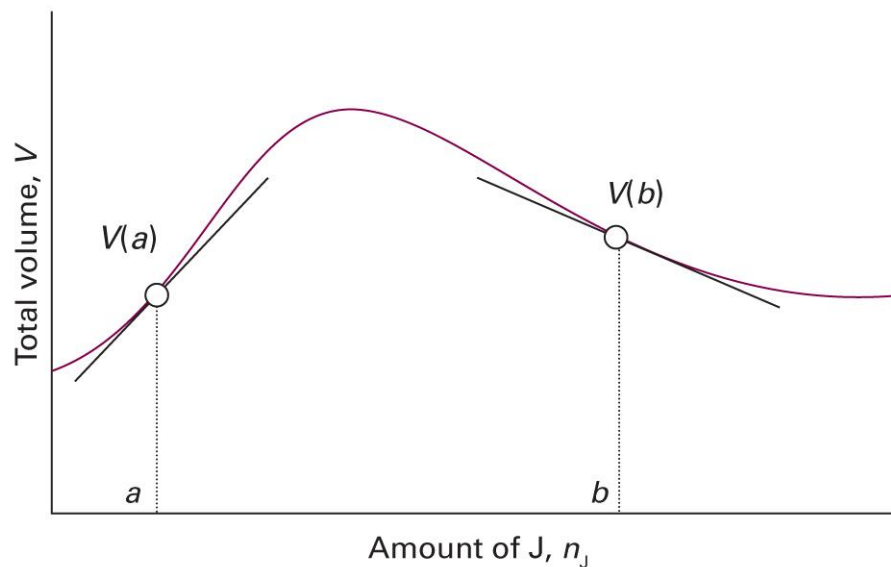
Partial Molar Quantities

$$X_J = \left(\frac{\partial X}{\partial n_J} \right)_{p,T,n'}$$

contribution that a component of a mixture makes to the total quantity of a sample

Partial Molar Volume

$$V_J = \left(\frac{\partial V}{\partial n_J} \right)_{p,T,n'}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5A.2)

Change of Volume

For a two-component system at constant p and T ,

$$V = V(n_A, n_B)$$

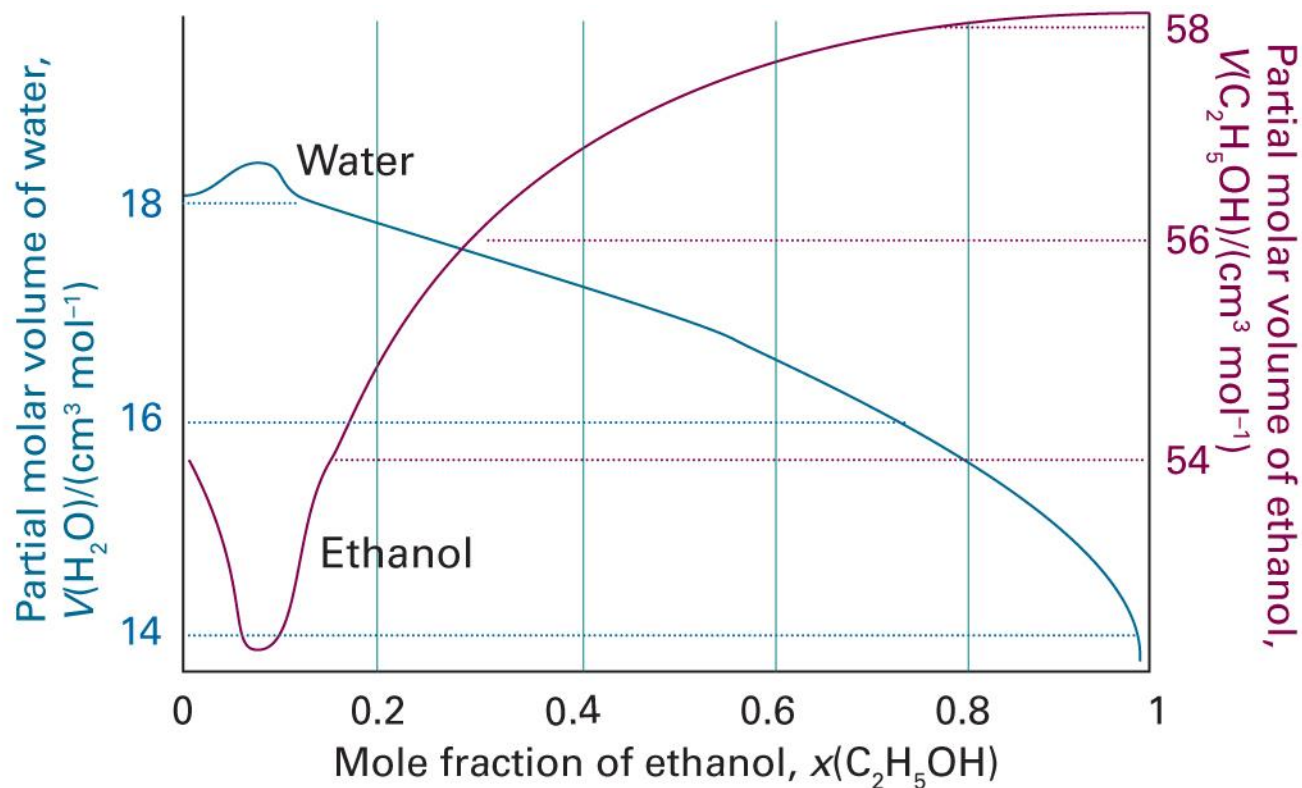
leads to

$$dV = \left(\frac{\partial V}{\partial n_A} \right)_{p,T,n_B} dn_A + \left(\frac{\partial V}{\partial n_B} \right)_{p,T,n_A} dn_B$$

Hence,

$$dV = V_A dn_A + V_B dn_B$$

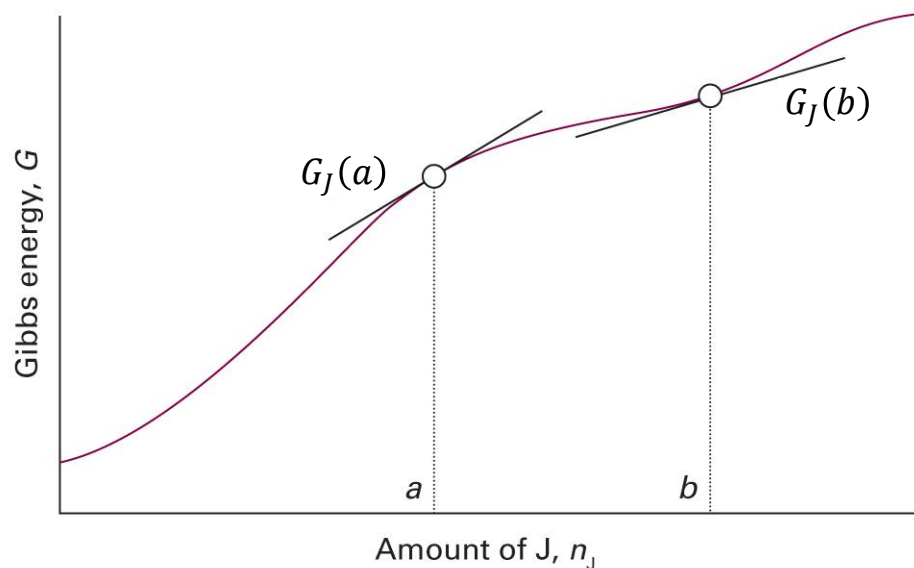
Example: Water + EtOH



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.1)

Partial Gibbs Energy

$$G_J = \left(\frac{\partial G}{\partial n_J} \right)_{p,T,n'}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5A.4)

... is Chemical Potential

$$dU = TdS - pdV + \mu_A dn_A + \mu_B dn_B + \dots$$

$$dH = TdS + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$$

$$dA = -SdT - pdV + \mu_A dn_A + \mu_B dn_B + \dots$$

$$dG = -SdT + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$$

Hence,

$$\mu_J = \left(\frac{\partial U}{\partial n_J} \right)_{S,V,n'} = \left(\frac{\partial H}{\partial n_J} \right)_{S,p,n'} = \left(\frac{\partial A}{\partial n_J} \right)_{T,V,n'} = \left(\frac{\partial G}{\partial n_J} \right)_{T,p,n'}$$

Be Careful

$$\mu_J = \left(\frac{\partial U}{\partial n_J} \right)_{S,V,n'} = \left(\frac{\partial H}{\partial n_J} \right)_{S,p,n'} = \left(\frac{\partial A}{\partial n_J} \right)_{T,V,n'} = \left(\frac{\partial G}{\partial n_J} \right)_{T,p,n'}$$

$$U_J = \left(\frac{\partial U}{\partial n_J} \right)_{T,p,n'} \neq \mu_J$$

$$H_J = \left(\frac{\partial H}{\partial n_J} \right)_{T,p,n'} \neq \mu_J$$

$$A_J = \left(\frac{\partial A}{\partial n_J} \right)_{T,p,n'} \neq \mu_J$$

Gibbs-Duhem Equation

Now, thermodynamic equations for internal energy are

$$U = TS - pV + \sum_J \mu_J n_J$$

$$dU = TdS - pdV + \sum_J \mu_J dn_J$$

Differentiating the first equation,

$$dU = TdS + SdT - pdV - Vdp + \sum_J \mu_J dn_J + \sum_J n_J d\mu_J$$

Comparison leads to

$$SdT - Vdp + \sum_J n_J d\mu_J = 0$$

At Constant T and p

$$SdT - Vdp + \sum_J n_J d\mu_J = 0$$

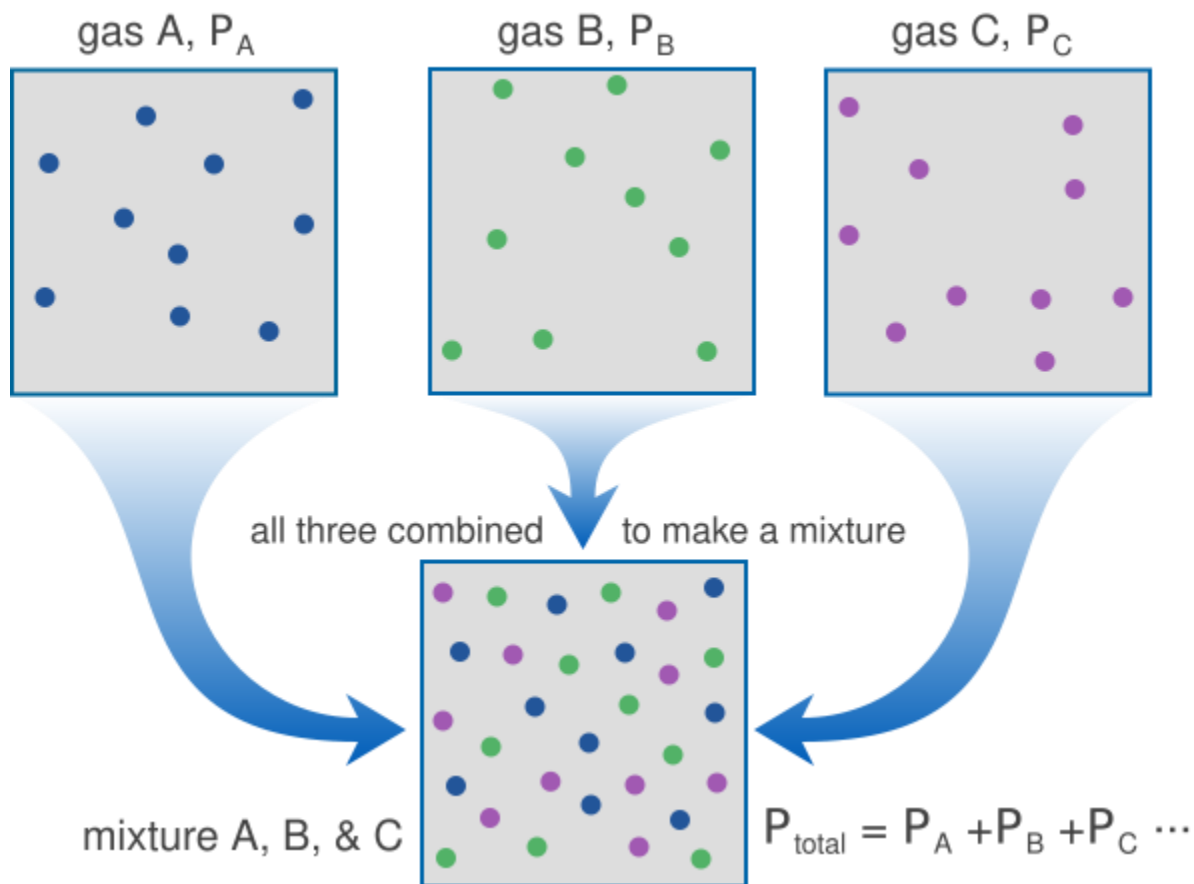
At constant T and p ,

$$\sum_J n_J d\mu_J = 0$$

For a two-component system,

$$n_A d\mu_A + n_B d\mu_B = 0$$

Gas Mixture



Dalton's Law

Partial pressure for species J is

$$p_J = x_J p$$

so

$$p_A + p_B + \cdots = (x_A + x_B + \cdots)p = p$$

Valid for all gases (not only perfect)

Chemical Potential

For a perfect gas,

$$G_m(p) = G_m^\ominus + RT \ln \frac{p}{p^\ominus}$$

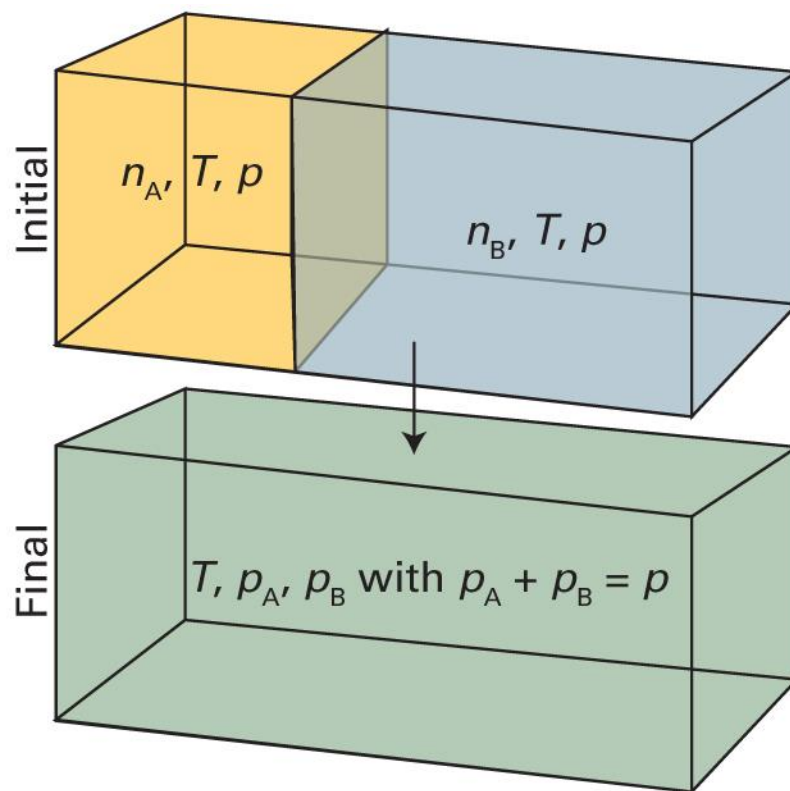
or,

$$\mu = \mu^\ominus + RT \ln \frac{p}{p^\ominus}$$

where $\mu^\ominus$ is the standard chemical potential (at 1 bar).

Mixing of Gases

(at constant T and p)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5A.6)

Total Gibbs Energy

$$G = G_A + G_B = n_A\mu_A + n_B\mu_B$$

$$\mu = \mu^\ominus + RT \ln(p/p^\ominus)$$

- Initial state

$$G_i = n_A\{\mu_A^\ominus + RT \ln(p/p^\ominus)\} + n_B\{\mu_B^\ominus + RT \ln(p/p^\ominus)\}$$

- Final state

$$G_f = n_A\{\mu_A^\ominus + RT \ln(p_A/p^\ominus)\} + n_B\{\mu_B^\ominus + RT \ln(p_B/p^\ominus)\}$$

Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = G_f - G_i = n_A RT \ln \left(\frac{p_A}{p} \right) + n_B RT \ln \left(\frac{p_B}{p} \right)$$

Using Dalton's law,

$$\Delta_{\text{mix}}G = n_A RT \ln x_A + n_B RT \ln x_B$$

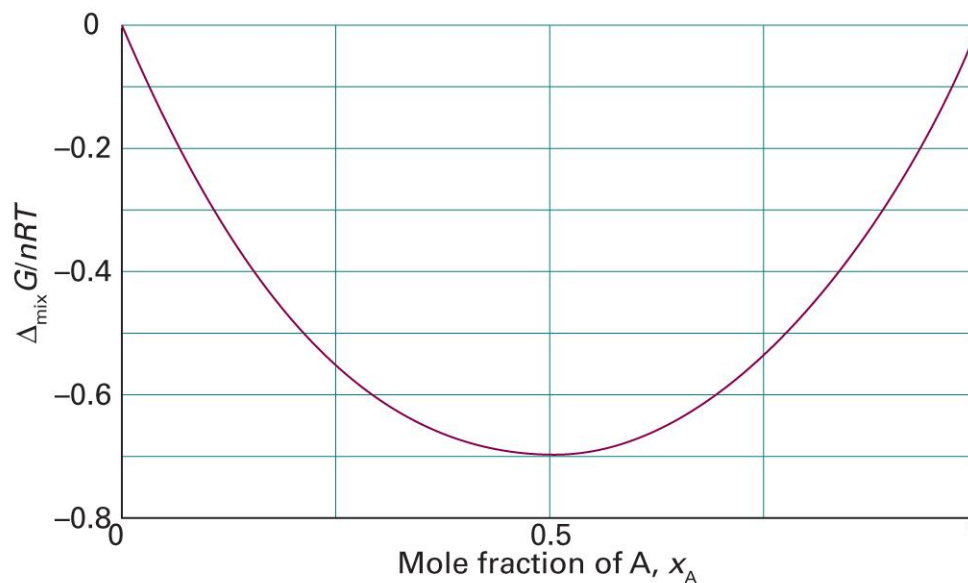
and

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

Note: since $\ln x < 0$ for $x < 1$, $\Delta_{\text{mix}}G < 0$ always.

Gibbs Energy of Mixing

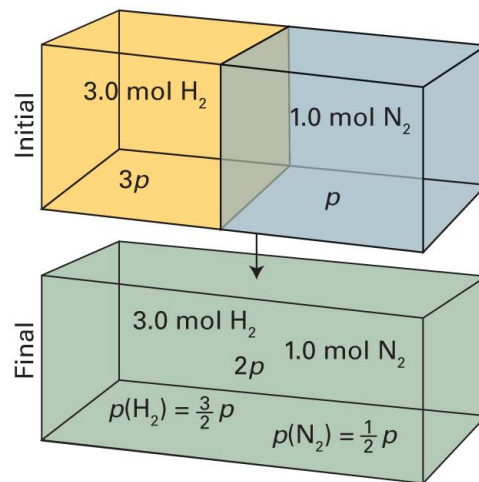
$$\Delta_{\text{mix}}G = nRT\{x_A \ln x_A + (1 - x_A) \ln(1 - x_A)\}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5A.7)

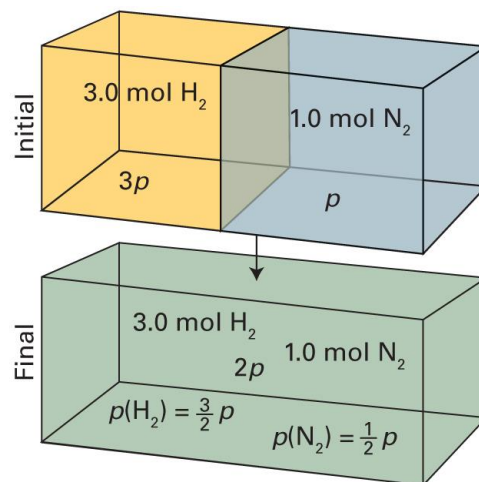
Example 5A.3

A container is divided into two equal compartments. One contains 3.0 mol $\text{H}_2(\text{g})$ at 25°C ; the other contains 1.0 mol $\text{N}_2(\text{g})$ at 25°C . Calculate the Gibbs energy of mixing when the partition is removed. Assume that the gases are perfect.



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5A.7)

Example 5A.3



- H₂: amount 3.0 mol, volume $V \rightarrow 2V$, pressure $3p \rightarrow (3/2)p$
- N₂: amount 1.0 mol, volume $V \rightarrow 2V$, pressure $p \rightarrow (1/2)p$

$$G = G_A + G_B = n_A \mu_A + n_B \mu_B$$

$$\mu = \mu^\ominus + RT \ln(p/p^\ominus)$$

Example 5A.3

$$G = G_A + G_B = n_A\mu_A + n_B\mu_B$$

$$\mu = \mu^\ominus + RT \ln(p/p^\ominus)$$

Total Gibbs energies are

$$G_i = n_A\{\mu_A^\ominus + RT \ln(3p/p^\ominus)\} + n_B\{\mu_B^\ominus + RT \ln(p/p^\ominus)\}$$

$$G_f = n_A\{\mu_A^\ominus + RT \ln(3p/2p^\ominus)\} + n_B\{\mu_B^\ominus + RT \ln(p/2p^\ominus)\}$$

Therefore,

$$\Delta_{\text{mix}}G = G_f - G_i = -n_A RT \ln 2 - n_B RT \ln 2 = -(n_A + n_B) RT \ln 2$$

$$= -(3.00 \text{ mol} + 1.00 \text{ mol}) \times (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K}) \times 0.693$$

$$= -6.9 \text{ kJ}$$



Entropy and Enthalpy

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

- Entropy of mixing

$$\Delta_{\text{mix}}S = -\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T}\right)_p = -nR(x_A \ln x_A + x_B \ln x_B)$$

- Enthalpy of mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T\Delta_{\text{mix}}S = 0$$

Real Gases

For a perfect gas,

$$\mu = \mu^\ominus + RT \ln \frac{p}{p^\ominus}$$

For a real gas,

$$\mu = \mu^\ominus + RT \ln \frac{f}{p^\ominus}$$

where f is the **fugacity**.

Properties of Fugacity

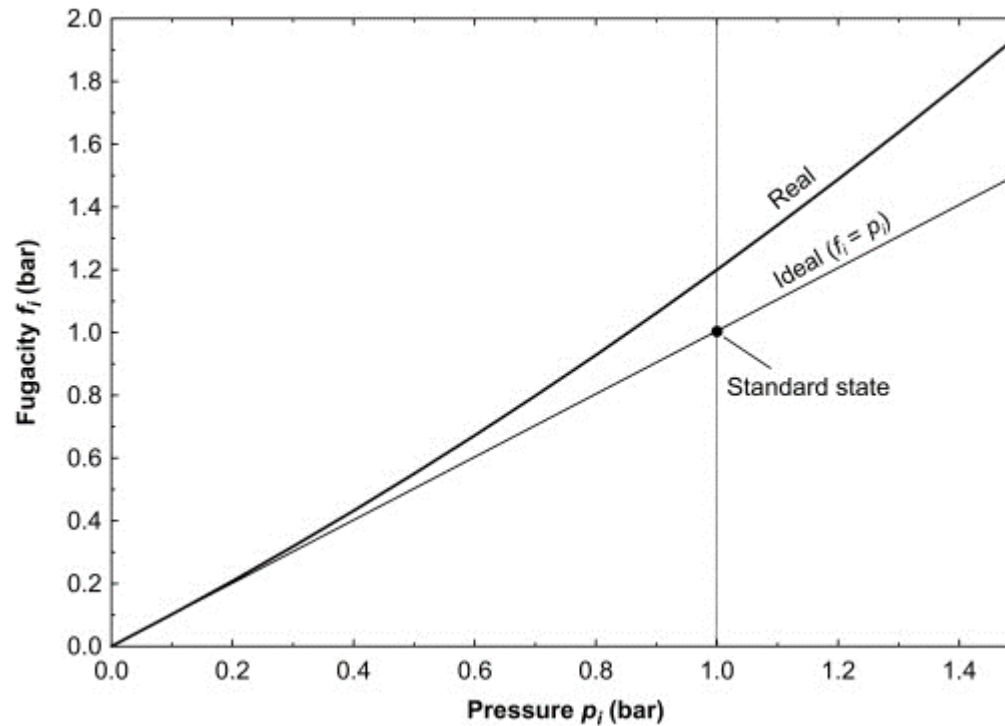
- As $p \rightarrow 0$, $f \rightarrow p$
- Fugacity coefficient $\phi = f/p$
- Molecular interpretation
 - $f > p$ or $\phi > 1$: repulsion-dominant
 - $f = p$ or $\phi = 1$: perfect-gas-like
 - $f < p$ or $\phi < 1$: attraction-dominant

NOTE

$$\begin{aligned}\mu &= \mu^\ominus + \int_{p^\ominus}^p V_m \, dp \\ &= \mu^\ominus + \int_{p^\ominus}^p \frac{RT}{p} \, dp \\ &= \mu^\ominus + RT \ln \frac{p}{p^\ominus}\end{aligned}$$

Not in Atkins

Fugacity and Pressure



Gibbs Energy: Before Mixing

$$G = G_A + G_B = n_A\mu_A + n_B\mu_B$$

$$\mu = \mu^\ominus + RT \ln \left\{ \frac{f(p)}{p^\ominus} \right\} = \mu^\ominus + RT \ln \left(\frac{p\phi}{p^\ominus} \right)$$

- Initial state: separated A and B gases at p and T

$$G_i = n_A \left[\mu_A^\ominus + RT \ln \left(\frac{p\phi_A^*}{p^\ominus} \right) \right] + n_B \left[\mu_B^\ominus + RT \ln \left(\frac{p\phi_B^*}{p^\ominus} \right) \right]$$

where ϕ_j^* denotes the fugacity coefficient of species J in its pure form.

Gibbs Energy: After Mixing

$$G = G_A + G_B = n_A\mu_A + n_B\mu_B$$

$$\mu = \mu^\ominus + RT \ln \left\{ \frac{f(p)}{p^\ominus} \right\} = \mu^\ominus + RT \ln \left(\frac{p\phi}{p^\ominus} \right)$$

- Final state: A and B gases in one container at p and T

From Dalton's law, $p_A = x_A p$ and $p_B = x_B p$

$$G_f = n_A \left[\mu_A^\ominus + RT \ln \left(\frac{x_A p \phi_A}{p^\ominus} \right) \right] + n_B \left[\mu_B^\ominus + RT \ln \left(\frac{x_B p \phi_B}{p^\ominus} \right) \right]$$

Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = G_f - G_i = n_A RT \ln \left(x_A \frac{\phi_A}{\phi_A^*} \right) + n_B RT \ln \left(x_B \frac{\phi_B}{\phi_B^*} \right)$$

$$\Delta_{\text{mix}}G = nRT \left\{ \underbrace{x_A \ln x_A + x_B \ln x_B}_{\text{perfect gas}} + \underbrace{x_A \ln \left(\frac{\phi_A}{\phi_A^*} \right) + x_B \ln \left(\frac{\phi_B}{\phi_B^*} \right)}_{\text{correction}} \right\}$$

In typical conditions, the **correction term** is much smaller and

$$\Delta_{\text{mix}}G = \Delta_{\text{mix}}H - T\Delta_{\text{mix}}S < 0$$